Day 46 — Technical Interview Flow Design

1. Objective

The objective is to define the flow and state transitions of a role-based technical interview session.

2. Interview Flow

The configured interview flow follows these states:

1. Introduction
2. Experience-based
3. Conceptual
4. Scenario-based
5. Evaluation
6. Closing
7. Completed

3. State Transition Model

Current State	Next State
Introduction	Experience-based
Experience-based	Conceptual
Conceptual	Scenario-based
Scenario-based	Evaluation
Evaluation	Closing
Closing	Completed

4. Interview Session Initialization

When a session starts, the engine:

1. Validates the selected role.
1. Validates the candidate's experience.
1. Determines the experience category.
1. Loads the configured question set.
1. Initializes the interview state.

1. Returns the first available question.

5. Response Capture

The `submit_response()` method:

1. Validates the answer type.
2. Retrieves the current question.
3. Stores the question, skill, difficulty, and candidate response.
4. Advances the question index.
5. Returns the next available question.

6. Evaluation Input

The `generate_evaluation_input()` method produces structured data containing:

1. Candidate role.
1. Experience classification.
1. Submitted questions.
1. Relevant technical skills.
1. Question difficulty.
1. Candidate answers.

This output can be used by a future technical scoring component.

7. State Management

The `advance_flow()` method uses the configured transitions to move between interview states.

The engine prevents transition to evaluation from the scenario-based state until all configured questions have received responses.

8. Testing and Validation

The Day 46 test suite validates:

1. Experience classification.
1. Role-to-skill mapping.
1. Interview section structure.

1. Interview initialization.
1. Question selection.
1. Response capture.
1. Evaluation input generation.
1. State transitions.
1. Invalid role and experience inputs.
1. Session reset behavior.

9. Limitations

The current implementation provides a configurable state flow and question response capture.

It does not yet provide automatic answer scoring, adaptive follow-up generation, or dynamic question difficulty changes.

These functions can be integrated in subsequent development tasks.